

43. CLASSIFICATION OF THE DIFFERENT METABOLIC FIELDS

DURATION : 18:39

STUDY SHEET

COURSE SUMMARY

This video explores the 'Classification of Different Metabolic Fields', focusing on two key concepts: the Corrosion Field and the Aspiration Field. The Corrosion Field is illustrated by the process of fusion of the primitive aortas, where the disappearance of internal tissues creates new structures, such as the branchial arches, and illustrates pathological phenomena such as bedsores. In parallel, the Aspiration Field describes how glands are formed through the aspiration and manipulation of epithelial tissues, taking examples such as the formation of the lungs and the diaphragm. This video provides an in-depth understanding of the metabolic transformations at work during embryonic development.

CLASSIFICATION OF DIFFERENT METABOLIC FIELDS

I. THE CORROSION FIELD: CREATION OF NEW STRUCTURES

A. FUSION OF PRIMITIVE AORTAS

Initially, we have **two primitive aortas** in the middle, with internal nutritive tissues for the developing embryo.

The influential **acneural** referent, which becomes a groove, creates a trajectory for the embryo. This trajectory initiates the formation of two small primitive aortic conduits.

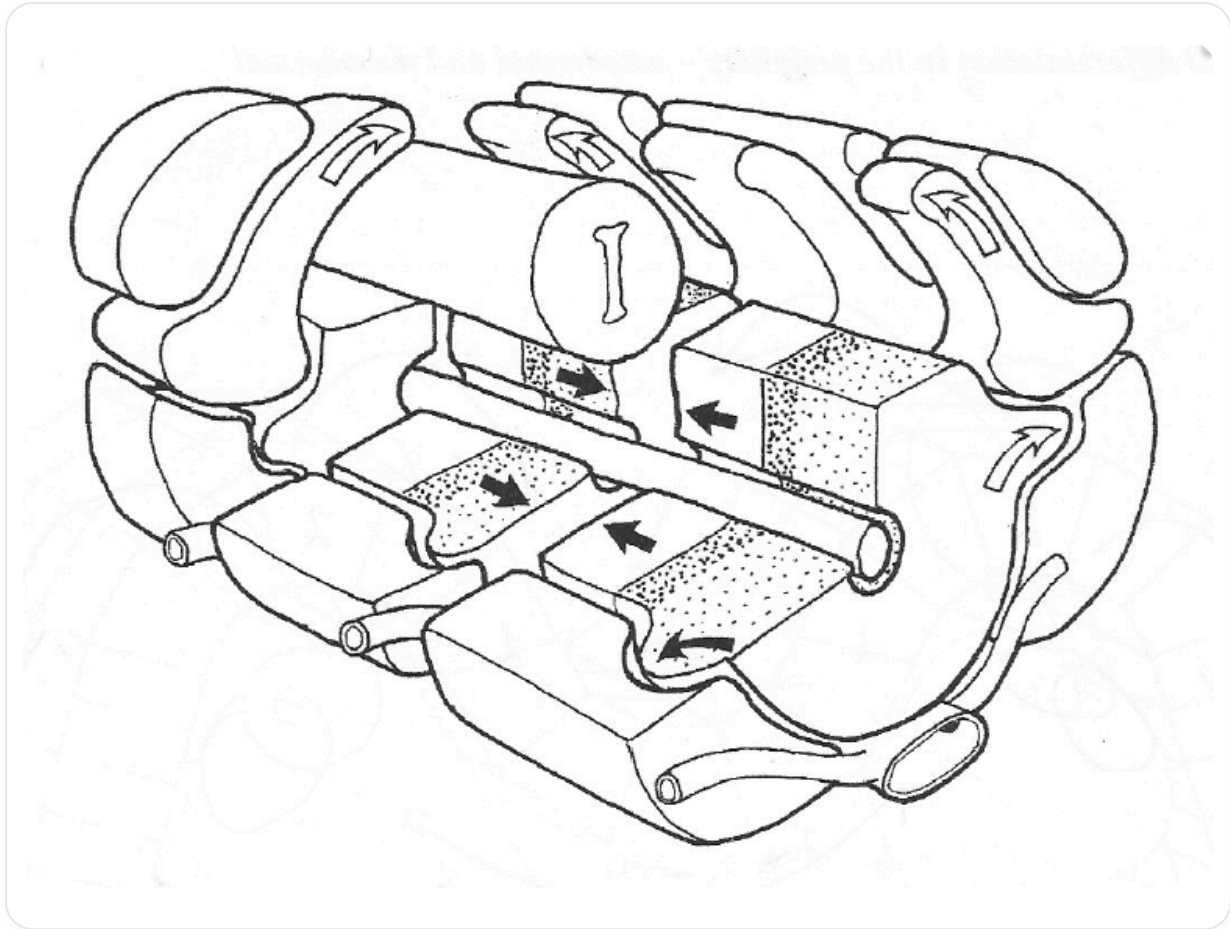


FIG. — Somites and neural tube

The embryo performs a movement of **flexion**, **internal rotation**, and **adduction**, a developmental movement where it moves towards the centre. This approximation is due to the development of the aortas and a process of **telencephalisation** which pulls backwards, bringing everything forwards.

During this phase, the two aortic tubes, initially distinct, converge towards the midline and eventually **fuse**.

B. THE FUSION PROCESS AND CORROSION

During this fusion, the internal tissues gradually diminish until they disappear.

The absence of these internal tissues creates a **corrosion field**. This is a common phenomenon in the body for the formation of various structures.

C. EXAMPLE OF BRANCHIAL ARCHES

A concrete example is the formation of the **branchial arches**. During embryonic flexion, compression appears externally, on the surface ectoderm, forming the branchial arches.

Pressure exerted by the branchial arch against the heart and mandible opens a space: the **mouth**. This is an example of a corrosion field.

D. CORROSION IN A PATHOLOGICAL CONTEXT: THE BEDSORE

Another example of a corrosion field, although pathological, is the **bedsore**. Prolonged hospitalisation in bed can cause constant pressure, poor blood supply, and a loss of trophic aspect.

This first leads to overheating, then to a corrosion field. This is the appearance of a new structure due to the destruction of internal tissue.

This concept of a corrosion field is fundamental in the development of the **vascular system**, where structures are constantly changing.

E. REORGANISATION OF THE VASCULAR SYSTEM

Initially, we have the **cardinal veins** (anterior and posterior), which then evolve into **subcardinal** veins, then **supracardinal** veins.

This complex reorganisation of the body depends on growth. Compression and decompression fields allow for the appearance of new structures, a process referred to as a corrosion field.

This is not destruction, but a **change in form**.

II. THE ASPIRATION FIELD (LOOSENING KILL): FORMATION OF GLANDS

A. GENERAL FUNCTIONING OF THE ASPIRATION FIELD

We will now address the "**loosening kill**", or aspiration and suction field, a fundamental mechanism for the formation of all **glands** in the body.

This type of field always appears along the surfaces of **epithelial** tissues, whether they are of **ectodermal** or **endodermal** origin.

B. EXAMPLE OF ENDODERMAL EPITHELIAL TISSUE

Let's take the example of **endodermal** epithelial tissue superimposed on connective tissue rich in vessels, lamps, matrix, and fibres.

Initially, a **congestive** phase is observed along the surfaces, like an accumulation of water. This is similar to walking on a beach where the ground appears congested but can actually give way under weight, like **quicksand**.

C. FORMATION OF THE LUNG AND DIAPHRAGM

Consider the **lung** and the establishment of the **diaphragm**. Behind the digestive tract, on the tissue, there is a weakening.

With the growth and lengthening of the embryo, a space opens, creating an aspiration field. The epithelial tissue is as if sucked into this space, like a suction cup.

The epithelial tissue descends into the connective tissue. It is aspirated and collapses into this connective tissue, forming an aspiration field around it.

D. DEVELOPMENT OF GLANDS: EXOCRINE AND ENDOCRINE

Following this aspiration, two types of cellular movements can occur:

1. **Exocrine glands:** The cells descend and form a gland that secretes a substance into a lumen created by a tube. For example, the **gallbladder** is an exocrine gland.
2. **Endocrine glands:** If the process goes further into the metabolic field, the gland becomes "swollen", disappears, and leaves an isolated gland in the middle. This secretes its substance directly into the **vascular** network.

The **pancreas** is an excellent example of this duality, with an exocrine pancreas and an endocrine pancreas. They originate from the same aspiration field, but some elements have lost their connection to become **cellular islets** producing hormones (insulin) or enzymes (amylase, lipase).

The lungs themselves can be considered a **respiratory gland**, following the same developmental algorithm but with different spaces and chronology. The **pleural vacuum** maintains their shape.

The difference between these glands is not in the initial metabolic field, but in the growth and loss of connection with the epithelial surface.

E. ENVIRONMENT AND GENETIC RESPONSE

Life begins as a vast aspiration field, where the environment interacts with the **genetic** response. The environment releases molecules (**morphogens**) that influence the tissue.

Corrosion fields create new structures, like the bedsore which is a destruction of internal tissue with the appearance of a new structure.

These embryonic processes, determined by the **genome**, space, and time, actively participate in the creation of the different metabolic fields.

III. DILATION, RETENTION, AND DETRACTION FIELDS

A. THE PRINCIPLE OF HOMEOSTASIS

Each metabolic field contributes to this creation, and disturbing one field can disrupt the entire system. This is the principle of **homeostatic guarantee**, where everyone must achieve their balance.

We are in a structure under construction. A **mesoblastic** tissue appears at a given moment, at a certain periphery.

B. FROM DILATION TO DETRACTION

Growth pulls on the tissue, which lengthens while retaining its elasticity by filling with **elastin**. This is a **dilation field**.

If the stretching continues and deepens, the tissue becomes a **retention field**. Initially perceived as a muscle, excessive tension transforms it into a **ligament** or **fascia**.

If the traction prolongs and intensifies further, we enter a **detraction field**. The fluid structure is lost, and the tissue eventually becomes **bone**.

Thus, a continuous dilation field gives a retention field, which in turn, by prolonging, can lead to a detraction field and form bone. This explains the appearance of bone in different ways (membranous or cartilaginous).

IV. CONTUSION AND DISTUSION FIELDS

A. UNDERSTANDING THE FIELDS

Tissue development is influenced by **contusion** and **distusion** fields.

These fields result either from the compression of cells, leading to the formation of **cartilaginous** tissues, or from a metabolic growth mechanism.

B. EFFECTS ON TISSUE THICKNESS

These fields lead to differences in tissue thickness. Each tissue reaches a balance according to its destiny, its bodily situation, its genome and its environment, as well as its growth force.

The role is to maintain this homeostatic principle through the primitive embryonic fields.

C. THE SACRUM AND THE VERTEBRAL AXIS

It is crucial to remember that the **sacrum** is at the base of cortical development.

If the sacrum is mobile, the **vertebral axis** is mobile. If the vertebral axis is mobile, the **medullary** aspect is mobile. If the medullary level is mobile, the **cranium** is mobile.

The sacrum is a structure to be systematically checked.

D. KEY EMBRYONIC SYNCHRONICITY

Synchronisation is essential. The neural tube, for example, articulates with the **cardiopulmonary** cavity.

The **urogenital** axis, by its mesodermal force, is the articulation between the endo and the ecto. The mesoderm is the articulation between the endoderm and the ectoderm.

It is necessary to integrate the **vascular system** to fully understand this scheme, highlighting the importance of the brain, liver, heart, lungs, mesenteric plexus, and renal system.

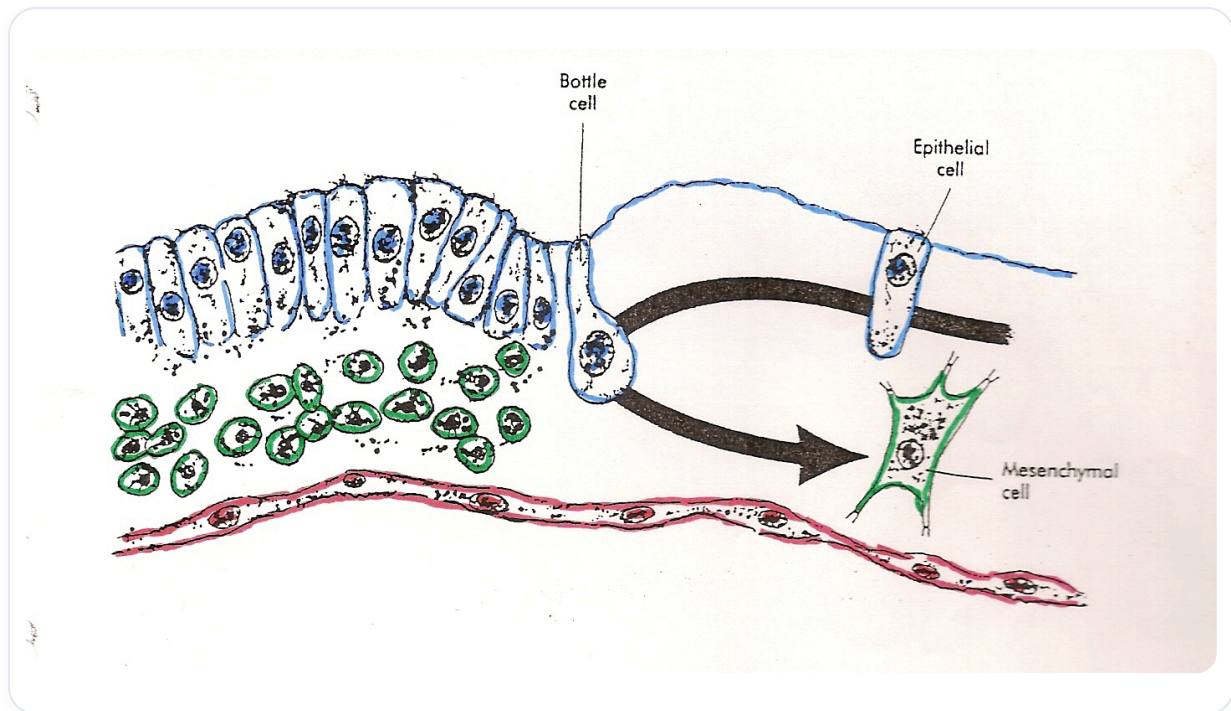


FIG. — EMT (Transition Épithélio-Mesenchymateuse)